argo-rdp.codeargo.net/web-rdp/#/client/aS0wODc5NTM0NmE2OGRiZDY4OQBjAGpzb24?token=C3DE77A17C8FC8582DBE3676F58B52B1E97F6EB9...

localhost:8891/notebooks/Downloads/ML_Assignment1_SIMPLE.ipynb?

jupyter ML_Assignment1_SIMPLE Last Checkpoint: 6 minutes ago

File Edit View Run Kernel Settings Help Trusted

Code JupyterLab Python [conda env:base] *

```
    y_val_best_predictions = y_val_predictions_poly
elif best_model_idx == 2:
    y_val_best_predictions = y_val_predictions_ridge
else:
    y_val_best_predictions = y_val_predictions_lasso

residuals = y_val - y_val_best_predictions

print(f"Using {best_model_name} for residuals (best validation RMSLE)")

# Create residual plots
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Plot 1: Predicted vs Actual
axes[0].scatter(y_val, y_val_best_predictions, alpha=0.5, color='steelblue')
axes[0].plot([y_val.min(), y_val.max()], [y_val.min(), y_val.max()],
             'r--', linewidth=2, label='Perfect Prediction')
axes[0].set_xlabel('Actual Values')
axes[0].set_ylabel('Predicted Values')
axes[0].set_title('Predicted vs Actual Values')
axes[0].legend()
axes[0].grid(alpha=0.3)

# Plot 2: Residual Distribution
axes[1].hist(residuals, bins=50, color='coral', edgecolor='black', alpha=0.7)
axes[1].axvline(x=0, color='red', linestyle='--', linewidth=2, label='Zero Error')
axes[1].set_xlabel('Residuals (Actual - Predicted)')
axes[1].set_ylabel('Frequency')
axes[1].set_title('Distribution of Residuals')
axes[1].legend()
axes[1].grid(alpha=0.3)
```

Anaconda Assistant

AI-powered coding, insights and debugging in your notebooks.

To enable the following extensions, create an account or sign in.

Anaconda Assistant 4.1.0

Coming soon!

Data Catalogs

Panel Deployments

Sharing

Create Account

Already have an account? Sign In

For more information, read our Anaconda Assistant documentation.

Q6. Try Advanced Models

Try:

1. Polynomial features (degree 2)
2. Ridge Regression (alpha=10.0)
3. Lasso Regression (alpha=0.1)

[26]:

```
print("="*70)
print("Q6: TRYING ADVANCED MODELS")
print("="*70)

# Store results
results = []

# Model 1: Simple Linear Regression (already done)
results.append({
    'Model': 'Linear Regression',
    'Features': 'Original',
    'Train RMSLE': train_error_rmsle,
    'Val RMSLE': val_error_rmsle
})

print("\n1. Trying Polynomial Features (degree 3)...")
# Create polynomial features (degree 3) for all features
poly = PolynomialFeatures(degree=3, include_bias=False)
X_train_poly = poly.fit_transform(X_train)
X_val_poly = poly.transform(X_val)
```

```
train_error_lasso = rmsle(y_train, y_train_predictions_lasso)
val_error_lasso = rmsle(y_val, y_val_predictions_lasso)

results.append({
    'Model': 'Lasso Regression',
    'Features': 'Polynomial (degree 3)',
    'Train RMSLE': train_error_lasso,
    'Val RMSLE': val_error_lasso
})

print(f"   Train RMSLE: {train_error_lasso:.4f}, Val RMSLE: {val_error_lasso:.4f}")
```

```
======================================================================
Q6: TRYING ADVANCED MODELS
======================================================================

1. Trying Polynomial Features (degree 3)...
   Train RMSLE: 1.2231, Val RMSLE: 1.2160

2. Trying Ridge Regression (alpha=10.0)...
   Train RMSLE: 1.2188, Val RMSLE: 1.1943

3. Trying Lasso Regression (alpha=0.1)...
   Train RMSLE: 1.2823, Val RMSLE: 1.2290
```

Answer to Q6:

Tried 3 approaches:

- Polynomial Features (degree 3) with Linear Regression
- Ridge Regression (α=10.0) with polynomial features
- Lasso Regression (α=0.1) with polynomial features

3D card flipping |...

Applications : Firefox Xfce Terminal

Welcome to Rocky Linux × Home × ML_Assignment1_SIMPLE × +

localhost:8891/notebooks/Downloads/ML_Assignment1_SIMPLE.ipynb?

Rocky Linux Rocky Wiki Rocky Forums Rocky Mattermost Rocky Reddit

jupyter ML_Assignment1_SIMPLE Last Checkpoint: 6 minutes ago

File Edit View Run Kernel Settings Help Trusted

Code JupyterLab Python [conda env:base] *

```python
plt.xticks(x, results_df['Model'], rotation=45, ha='right')
plt.legend()
plt.grid(alpha=0.3, axis='y')
plt.tight_layout()
plt.show()
```

```
======================================================================
Q7: MODEL COMPARISON TABLE
======================================================================

             Model              Features  Train RMSLE  Val RMSLE
Linear Regression               Original     1.329423   1.288430
Linear Reg (Poly) Polynomial (degree 3)     1.223057   1.215965
 Ridge Regression Polynomial (degree 3)     1.218800   1.194314
 Lasso Regression Polynomial (degree 3)     1.282340   1.229044

======================================================================
BEST MODEL: Ridge Regression
Validation RMSLE: 1.1943
======================================================================
```

Chrome File Edit View History Bookmarks Profiles Tab Window Help   Wed 27 May 4:35 AM

argo-rdp.codeargo.net/web-rdp/#/client/aS0wODc5NTM0NmE2OGRiZDY4OQBjAGpzb24?token=C3DE77A17C8FC8582DBE3676F58B52B1E97F6EB9...

3D card flipping |...   All Bookmarks

Applications Firefox Xfce Terminal   2026-05-27 04:35 Cloud User

Welcome to Rocky Linux | Home | ML_Assignment1_SIMPLE

localhost:8891/notebooks/Downloads/ML_Assignment1_SIMPLE.ipynb?

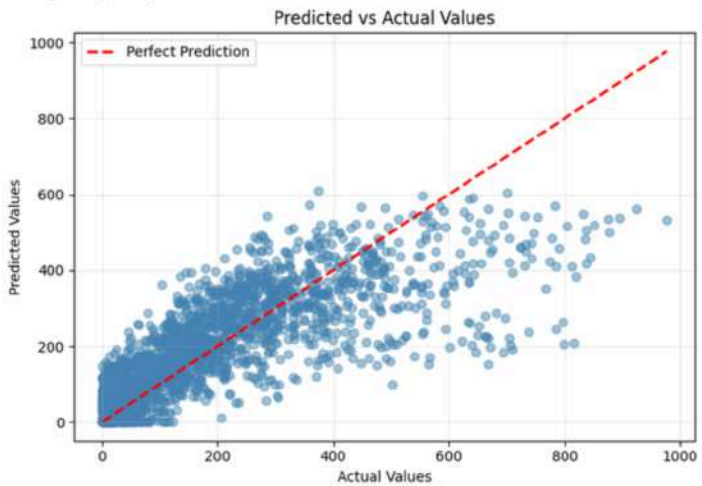
Rocky Linux Rocky Wiki Rocky Forums Rocky Mattermost Rocky Reddit

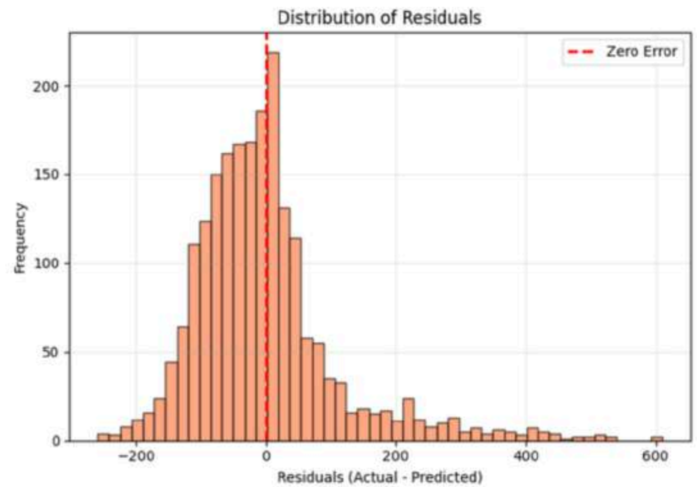
jupyter ML_Assignment1_SIMPLE Last Checkpoint: 6 minutes ago

File Edit View Run Kernel Settings Help   Trusted

Code   JupyterLab   Python [conda env:base] *

```
Using Ridge Regression for residuals (best validation RMSLE)
```






```
======================================================================
Q8: RESIDUALS
======================================================================

Residual Statistics:
  Mean: -2.75 (should be close to 0)
  Std Dev: 112.12
  Min: -258.68
  Max: 610.02
```

Anaconda Assistant

AI-powered coding, insights and debugging in your notebooks.

To enable the following extensions, create an account or sign in.

Anaconda Assistant 4.1.0

Coming soon!

Data Catalogs

Panel Deployments

Sharing

Create Account

Already have an account? Sign In

For more information, read our Anaconda Assistant documentation.

Need to find the balance - use validation error to pick the right complexity level.

Q12. Why can't Linear Regression alone capture time-of-day effects ly?

Answer to Q12:

Why Linear Regression doesn't work well with time-of-day:

Linear Regression assumes a straight line relationship - if hour increases, rentals increase or decrease at a constant rate.

But bike rentals have a cyclical daily pattern:

- Low at night (3am)
- Peak in morning (8am) - people going to work
- Medium at midday
- Peak in evening (5-6pm) - people coming back
- Low at night again

A straight line can't capture this up-and-down pattern with two peaks.

Hour 23 (11pm) and hour 0 (midnight) are next to each other in time, but numerically they're far apart (23 vs 0). Linear regression doesn't understand this cyclical nature.

Solution: Use polynomial features (like hour²) or interactions to capture these curved, non-linear patterns.

```
    Formula: sqrt(mean((log(pred+1) - log(actual+1))^2))
    """
    return np.sqrt(np.mean((np.log1p(y_pred) - np.log1p(y_true))**2))
```

Step 3: Load Data

[7]:
```
# Load training and test data
df_train = pd.read_csv('bike_train.csv')
df_test = pd.read_csv('bike_test.csv')

print(f"Training data shape: {df_train.shape}")
print(f"Test data shape: {df_test.shape}")
```

```
Training data shape: (10450, 12)
Test data shape: (2613, 9)
```

Q1. Examine dataset size, missing values, and feature types

[9]:
```
print("=" * 70)
print("Q1: DATASET OVERVIEW")
print("=" * 70)

# 1. Dataset Size
print(f"\n1. DATASET SIZE:")
print(f"   Rows: {df_train.shape[0]}")
print(f"   Columns: {df_train.shape[1]}")
```

3D card flipping |...

Applications Firefox Xfce Terminal

Welcome to Rocky Linux × Home × ML_Assignment1_SIMPLE × +

localhost:8891/notebooks/Downloads/ML_Assignment1_SIMPLE.ipynb?

Rocky Linux Rocky Wiki Rocky Forums Rocky Mattermost Rocky Reddit

jupyter ML_Assignment1_SIMPLE Last Checkpoint: 4 minutes ago

File Edit View Run Kernel Settings Help Trusted

Code JupyterLab Python [conda env:base] *

```python
print(df_train.head())

# 5. Basic statistics
print(f"\n5. BASIC STATISTICS:")
print(df_train.describe())
```

```
======================================================================
Q1: DATASET OVERVIEW
======================================================================

1. DATASET SIZE:
   Rows: 10450
   Columns: 12

2. MISSING VALUES:
   No missing values

3. FEATURE TYPES:
datetime       object
season          int64
holiday         int64
workingday      int64
weather         int64
temp          float64
atemp         float64
humidity        int64
windspeed     float64
casual          int64
registered      int64
count           int64
dtype: object

4. SAMPLE DATA:
```

3D card flipping |...

Applications Firefox Xfce Terminal

Welcome to Rocky Linux × Home × ML_Assignment1_SIMPLE × +

localhost:8891/notebooks/Downloads/ML_Assignment1_SIMPLE.ipynb?

Rocky Linux Rocky Wiki Rocky Forums Rocky Mattermost Rocky Reddit

jupyter ML_Assignment1_SIMPLE Last Checkpoint: 5 minutes ago

File Edit View Run Kernel Settings Help Trusted

Code JupyterLab Python [conda env:base] *

```
dtype: object

4. SAMPLE DATA:
             datetime  season  holiday  workingday  weather      temp  \
0   2012-07-15 7:00:00       3        0           0        1  28.70000
1  2012-08-14 15:00:00       3        0           1        1  33.62000
2   2011-02-06 6:00:00       1        0           0        1  10.66000
3  2012-05-06 17:00:02       2        0           0        2  26.42506
4   2012-01-09 2:00:00       1        0           1        1   9.84000

       atemp  humidity  windspeed  casual  registered  count
0  33.335000        79   6.003200      17          30     47
1  37.880000        46  15.001300      84         199    283
2  12.880000        60  15.001300       0           1      1
3  30.566166        61   9.512288     198         330    531
4  12.120000        56   8.998100       2           3      5

5. BASIC STATISTICS:
             season       holiday    workingday       weather          temp  \
count  10450.000000  10450.000000  10450.000000  10450.000000  10450.000000
mean       2.507943      0.028804      0.675694      1.413876     20.191700
std        1.116946      0.167263      0.468137      0.632258      7.792683
min        1.000000      0.000000      0.000000      1.000000      0.820000
25%        2.000000      0.000000      0.000000      1.000000     13.940000
50%        3.000000      0.000000      1.000000      1.000000     20.500000
75%        4.000000      0.000000      1.000000      2.000000     26.240000
max        4.000000      1.000000      1.000000      4.000000     41.000000

              atemp      humidity     windspeed        casual    registered  \
count  10450.000000  10450.000000  10450.000000  10450.000000  10450.000000
mean      23.605793     61.924211     12.765259     35.869091    154.511675
std        8.478045     19.245193      8.102821     49.629436    150.861267
min        0.760000      0.000000      0.000000      0.000000      0.000000
```

```
min        1.000000
25%       42.000000
50%      143.000000
75%      283.000000
max      977.000000
```

Answer to Q1:

Findings:

1. **Dataset Size:** 10,450 rows and 12 columns
2. **Missing Values:** No missing values - data is clean and ready to use
3. **Data Types and Feature Types:**
   - **Categorical:** season, holiday, workingday, weather
   - **Numerical:** temp, atemp, humidity, windspeed
   - **Datetime:** datetime column (needs processing)
   - **Target:** count (what we want to predict)
   - **Additional:** casual, registered (components of count)
4. **Target Variable:** `count` = `casual` + `registered`

Q2. Visualize relationships between features and target variable

[12]:
```
# First, extract hour from datetime for analysis
df_train['hour'] = pd.to_datetime(df_train['datetime'], format='mixed', dayfirst=True).dt.hour

# Create visualizations
fig, axes = plt.subplots(2, 2, figsize=(14, 10))
```

- **Datetime:** datetime column (needs processing)
- **Target:** count (what we want to predict)
- **Additional:** casual, registered (components of count)

4. **Target Variable:** `count` = `casual` + `registered`

Q2. Visualize relationships between features and target variable

[12]:

```
# First, extract hour from datetime for analysis
df_train['hour'] = pd.to_datetime(df_train['datetime'], format='mixed', dayfirst=True).dt.hour

# Create visualizations
fig, axes = plt.subplots(2, 2, figsize=(14, 10))

# Plot 1: Distribution of bike rentals
axes[0, 0].hist(df_train['count'], bins=40, color='steelblue', edgecolor='black')
axes[0, 0].set_xlabel('Bike Count')
axes[0, 0].set_ylabel('Frequency')
axes[0, 0].set_title('Distribution of Bike Rentals')
axes[0, 0].grid(alpha=0.3)

# Plot 2: Average rentals by hour
hourly_avg = df_train.groupby('hour')['count'].mean()
axes[0, 1].plot(hourly_avg.index, hourly_avg.values, marker='o', linewidth=2, color='darkgreen')
axes[0, 1].set_xlabel('Hour of Day')
axes[0, 1].set_ylabel('Average Rentals')
axes[0, 1].set_title('Average Bike Rentals by Hour')
axes[0, 1].grid(alpha=0.3)

# Plot 3: Temperature vs Rentals
```

3D card flipping |...

Applications Firefox Xfce Terminal

Welcome to Rocky Linux × Home × ML_Assignment1_SIMPLE ×

localhost:8891/notebooks/Downloads/ML_Assignment1_SIMPLE.ipynb?

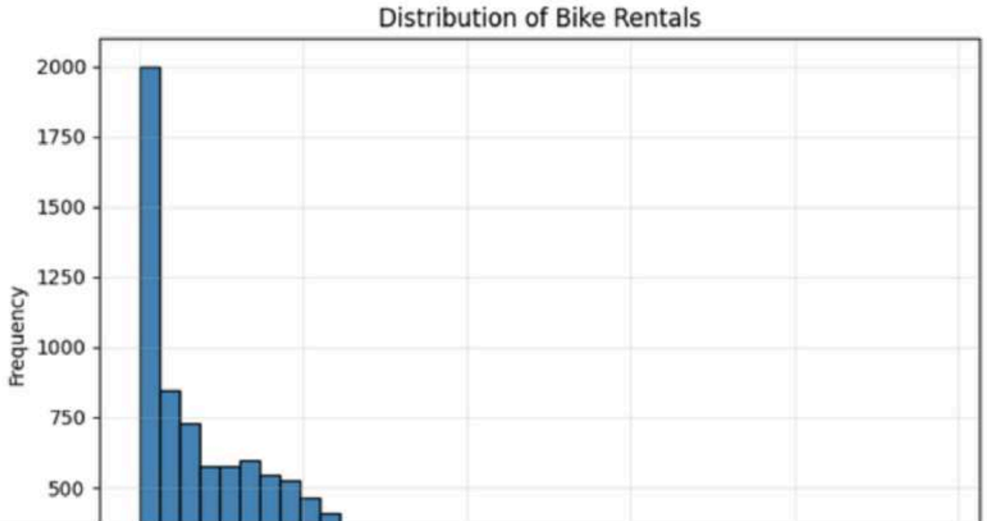
Rocky Linux Rocky Wiki Rocky Forums Rocky Mattermost Rocky Reddit

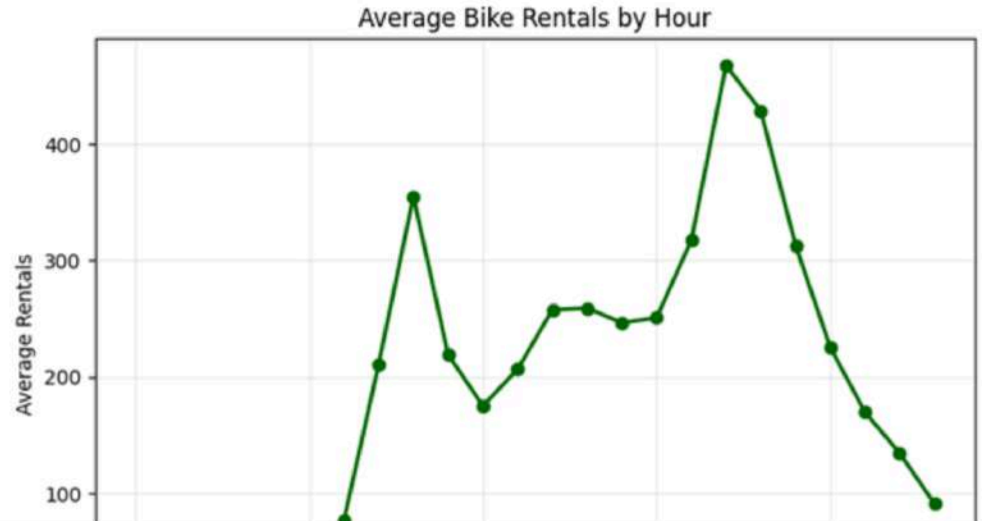
jupyter ML_Assignment1_SIMPLE Last Checkpoint: 5 minutes ago

File Edit View Run Kernel Settings Help Trusted

Code JupyterLab Python [conda env:base] *

```python
axes[1, 1].bar(weather_avg.index, weather_avg.values, color='purple', alpha=0.7)
axes[1, 1].set_xlabel('Weather (1=Clear, 2=Mist, 3=Light Snow/Rain, 4=Heavy Rain)')
axes[1, 1].set_ylabel('Average Rentals')
axes[1, 1].set_title('Average Rentals by Weather Condition')
axes[1, 1].grid(alpha=0.3)

plt.tight_layout()
plt.show()

# Print summary statistics
print("\nKey Statistics:")
print(f"Average hourly rentals: {df_train['count'].mean():.2f}")
print(f"Maximum hourly rentals: {df_train['count'].max()}")
print(f"Minimum hourly rentals: {df_train['count'].min()}")
```





3D card flipping |...

Applications Firefox Xfce Terminal

Welcome to Rocky Linux × Home × ML_Assignment1_SIMPLE × +

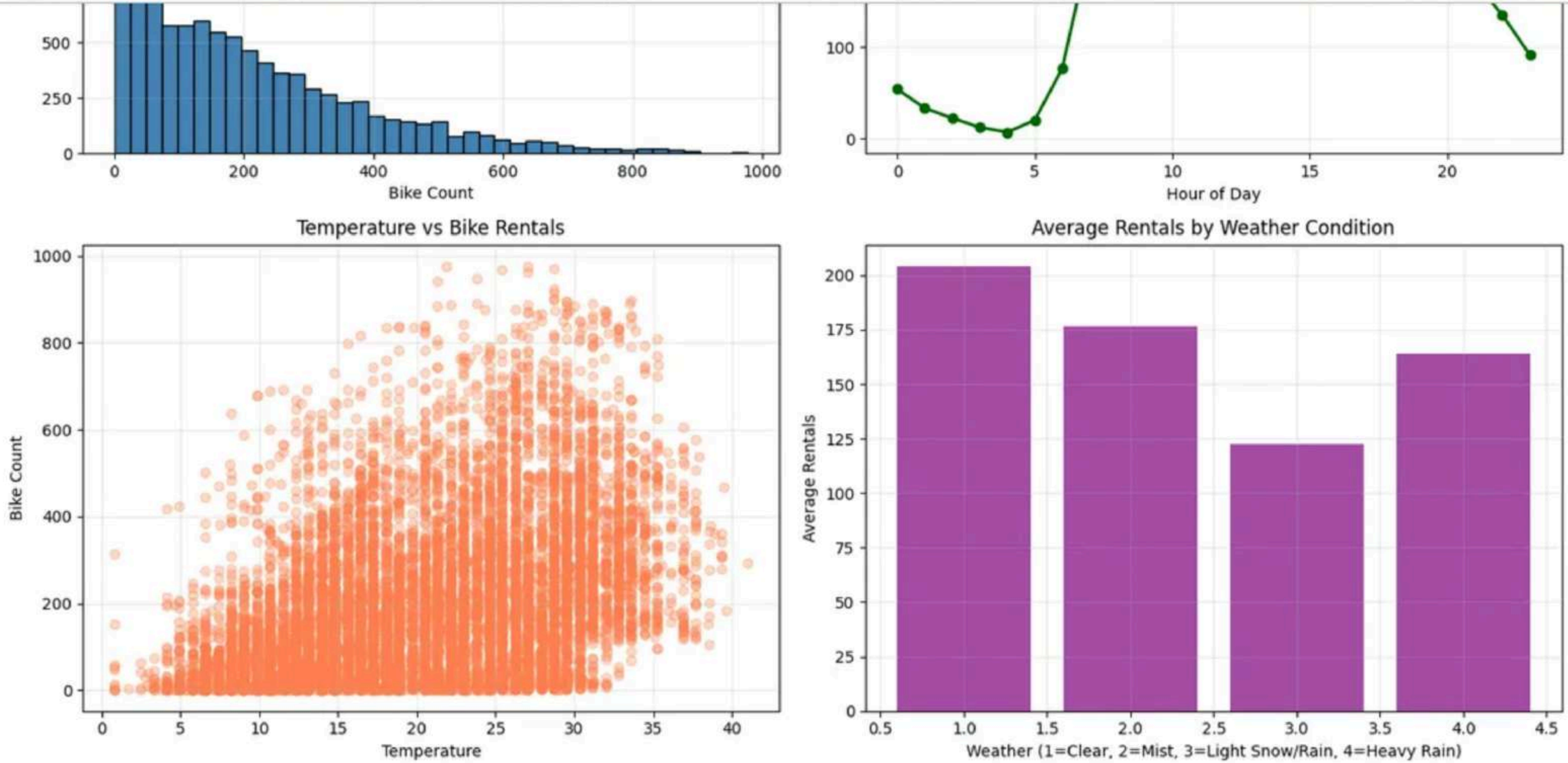
localhost:8891/notebooks/Downloads/ML_Assignment1_SIMPLE.ipynb?

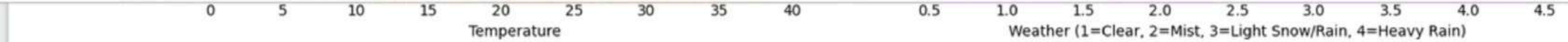
Rocky Linux Rocky Wiki Rocky Forums Rocky Mattermost Rocky Reddit

jupyter ML_Assignment1_SIMPLE Last Checkpoint: 5 minutes ago

File Edit View Run Kernel Settings Help Trusted

Code JupyterLab Python [conda env:base] *






```
Key Statistics:
Average hourly rentals: 190.41
Maximum hourly rentals: 977
Minimum hourly rentals: 1
```

Answer to Q2:

Observations from the plots:

From the plots above:

1. **Distribution of Rentals (Plot 1):**
   - Most times there are low to medium rentals
   - The graph looks skewed to the right (long tail)
2. **Time Patterns - Hour of Day (Plot 2):**
   - **Peak times:** 8am and 5-6pm (people going to work/coming back)
   - **Low rentals:** Late night and early morning (3-5am)
   - There's a daily pattern
3. **Temperature Effect (Plot 3):**
   - Higher temperature means more rentals
   - Most rentals happen between 15-30°C
   - Too cold or too hot = fewer rentals

- **Bad weather (3-4):** Much fewer rentals
- Weather matters a lot for bike rentals

Q3. Suggest which variables are likely to be most informative

[15]:

```python
# Calculate correlation with target variable
# First create a simple dataframe with numeric features and hour
df_train['month'] = pd.to_datetime(df_train['datetime'], format='mixed', dayfirst=True).dt.month
df_train['dayofweek'] = pd.to_datetime(df_train['datetime'], format='mixed', dayfirst=True).dt.dayofweek

# Select numeric features for correlation
numeric_features = ['season', 'holiday', 'workingday', 'weather',
                    'temp', 'atemp', 'humidity', 'windspeed',
                    'hour', 'month', 'dayofweek', 'count']

# Calculate correlations with target
correlations = df_train[numeric_features].corr()['count'].sort_values(ascending=False)

print("=" * 70)
print("Q3: FEATURE IMPORTANCE (Correlation with Target)")
print("=" * 70)
print("\nCorrelation with 'count' (target variable):\n")
print(correlations)

# Visualize correlations
plt.figure(figsize=(10, 6))
correlations[correlations.index != 'count'].plot(kind='barh', color='teal')
plt.xlabel('Correlation with Bike Count')
plt.ylabel('Features')
```

3D card flipping |...

Applications Firefox Xfce Terminal

Welcome to Rocky Linux × Home × ML_Assignment1_SIMPLE × +

localhost:8891/notebooks/Downloads/ML_Assignment1_SIMPLE.ipynb?

Rocky Linux Rocky Wiki Rocky Forums Rocky Mattermost Rocky Reddit

jupyter ML_Assignment1_SIMPLE Last Checkpoint: 5 minutes ago

File Edit View Run Kernel Settings Help Trusted

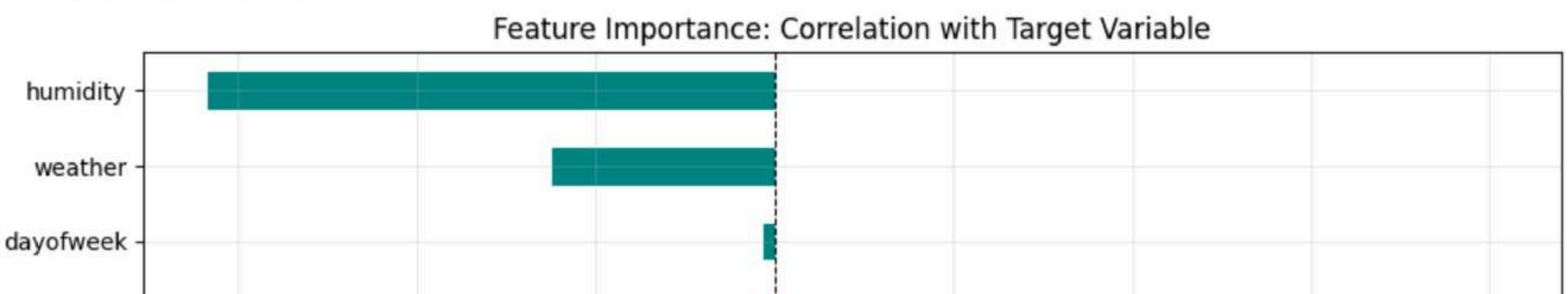
Code JupyterLab Python [conda env:base] *

```
print("=" * 70)
print("\nCorrelation with 'count' (target variable):\n")
print(correlations)

# Visualize correlations
plt.figure(figsize=(10, 6))
correlations[correlations.index != 'count'].plot(kind='barh', color='teal')
plt.xlabel('Correlation with Bike Count')
plt.ylabel('Features')
plt.title('Feature Importance: Correlation with Target Variable')
plt.axvline(x=0, color='black', linestyle='--', linewidth=0.8)
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```

```
======================================================================
Q3: FEATURE IMPORTANCE (Correlation with Target)
======================================================================

Correlation with 'count' (target variable):

count         1.000000
hour          0.404188
temp          0.396451
atemp         0.390642
month         0.165475
season        0.160333
windspeed     0.105318
workingday    0.017361
holiday      -0.005615
dayofweek    -0.006489
weather      -0.124402
humidity     -0.316607
```

```
plt.tight_layout()
plt.show()
```

```
======================================================================
Q3: FEATURE IMPORTANCE (Correlation with Target)
======================================================================

Correlation with 'count' (target variable):

count         1.000000
hour          0.404188
temp          0.396451
atemp         0.390642
month         0.165475
season        0.160333
windspeed     0.105318
workingday    0.017361
holiday      -0.005615
dayofweek    -0.006489
weather      -0.124402
humidity     -0.316607
Name: count, dtype: float64
```



−0.3 −0.2 −0.1 0.0 0.1 0.2 0.3 0.4
Correlation with Bike Count

Answer to Q3:

Feature Importance:

Strong Correlations:

1. **hour** (0.40) - Time of day matters a lot (rush hours)
2. **temp** (0.40) - Temperature affects how many people rent bikes
3. **atemp** (0.39) - "Feels like" temperature (almost same as temp)

Medium Correlations: 4. **season** (0.16-0.26) - Different seasons have different patterns 5. **month** (0.17) - Monthly patterns

Negative Correlations: 6. **humidity** (-0.32) - High humidity means less rentals 7. **weather** (-0.12) - Bad weather means less rentals

Weak/No Correlation:

- holiday, workingday, dayofweek (0.0 to ±0.02) - Don't help much by themselves

For building the model: Best to focus on **hour**, **temp**, **atemp**, and **humidity** because these have the strongest correlation with bike rentals. Can try combining **hour** with other features later.

Q4. Feature Engineering

3D card flipping |...

Applications Firefox Xfce Terminal

Welcome to Rocky Linux × Home × ML_Assignment1_SIMPLE × +

localhost:8891/notebooks/Downloads/ML_Assignment1_SIMPLE.ipynb?

Rocky Linux Rocky Wiki Rocky Forums Rocky Mattermost Rocky Reddit

jupyter ML_Assignment1_SIMPLE Last Checkpoint: 6 minutes ago

File Edit View Run Kernel Settings Help Trusted

Code JupyterLab Python [conda env:base] *

```
y_val_predictions = lr_simple.predict(X_val)

# Ensure non-negative predictions (rentals can't be negative)
y_train_predictions = np.maximum(0, y_train_predictions)
y_val_predictions = np.maximum(0, y_val_predictions)

# Calculate RMSLE
train_error_rmsle = rmsle(y_train, y_train_predictions)
val_error_rmsle = rmsle(y_val, y_val_predictions)

print("\n" + "="*70)
print("Q5: SIMPLE LINEAR REGRESSION RESULTS")
print("="*70)
print(f"\nTraining RMSLE: {train_error_rmsle:.4f}")
print(f"Validation RMSLE: {val_error_rmsle:.4f}")
```

```
Training set: 8360 samples
Validation set: 2090 samples

======================================================================
Q5: SIMPLE LINEAR REGRESSION RESULTS
======================================================================

Training RMSLE: 1.3294
Validation RMSLE: 1.2884
```

Answer to Q5:

Linear Regression Model:

- Used 80-20 train-validation split
- Features: 12 basic features (season, weather, temp, hour, etc.)

Machine Learning Assignment 1

Bike Sharing Demand Prediction

Student Name: Prateek srivastava

Date: May 26, 2026

Objective

Predict hourly bike rentals using Linear Regression.

Target Variable: `count` = total bike rentals (casual + registered)

Step 1: Import Libraries

[3]:

```
# Import necessary libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```